4_25

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Exercise 4.25: Housing Price Analysis in Collegetown

```
# Load necessary libraries
library(POE5Rdata)
library(tseries)
library(knitr)
library(ggplot2)

# Load data
data(collegetown)
summary(collegetown)
```

```
##      price             sqft            age             pool
##  Min.   :  50.0   Min.   :10.00   Min.   : 1.00   Min.   :0.000
##  1st Qu.: 157.9   1st Qu.:20.56   1st Qu.: 7.00   1st Qu.:0.000
##  Median : 199.9   Median :25.55   Median : 9.00   Median :0.000
##  Mean   : 250.2   Mean   :27.28   Mean   : 8.19   Mean   :0.064
##  3rd Qu.: 279.0   3rd Qu.:31.20   3rd Qu.:10.00   3rd Qu.:0.000
##  Max.   :1370.0   Max.   :91.67   Max.   :11.00   Max.   :1.000
##    fireplace         close          twostory        occupied
##  Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :0.00
##  1st Qu.:0.000   1st Qu.:0.000   1st Qu.:0.000   1st Qu.:0.00
##  Median :1.000   Median :0.000   Median :0.000   Median :1.00
##  Mean   :0.546   Mean   :0.378   Mean   :0.138   Mean   :0.56
##  3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:0.000   3rd Qu.:1.00
##  Max.   :1.000   Max.   :1.000   Max.   :1.000   Max.   :1.00
```

```
# Calculate sample means for elasticity and slope calculations
avg_sqft <- mean(collegetown$sqft)
avg_price <- mean(collegetown$price)
```

(a) Log-Linear Model

```
mod_a <- lm(log(price) ~ sqft, data = collegetown)
summary(mod_a)
```

```
##
## Call:
## lm(formula = log(price) ~ sqft, data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.32528 -0.19199  0.00122  0.21678  0.86609
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.393866   0.043288  101.50   <2e-16 ***
## sqft        0.036044   0.001486   24.26   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5408
## F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
b2 <- coef(mod_a)["sqft"]

# Calculating slope and elasticity at sample means
# Slope: dP/dx = beta2 * P
# Elasticity: epsilon = beta2 * x
slope_a <- b2 * avg_price
elasticity_a <- b2 * avg_sqft
```

Interpretation: The estimated coefficient $\hat{\beta}_2$ is approximately 0.036. This suggests that for every additional 100 square feet, the house price is expected to increase by approximately 3.6%. At the sample means, the marginal effect (slope) is 9.02 and the elasticity is 0.98.

(b) Log-Log Model

```
mod_b <- lm(log(price) ~ log(sqft), data = collegetown)
summary(mod_b)
```

```
##
## Call:
## lm(formula = log(price) ~ log(sqft), data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36864 -0.20849  0.00487  0.23204  1.21099
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.04965    0.15797   12.97   <2e-16 ***
## log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 498 degrees of freedom
## Multiple R-squared:  0.4738, Adjusted R-squared:  0.4728
## F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
a2 <- coef(mod_b)["log(sqft)"]

# Slope and Elasticity
# Slope: dP/dx = alpha2 * (P/x)
# Elasticity: alpha2
slope_b <- a2 * (avg_price / avg_sqft)
elasticity_b <- a2
```

Interpretation: The coefficient $\hat{\alpha}_2$ is 1.02, which represents a constant elasticity. A 1% increase in square footage is associated with a 1.02% increase in house price. At the sample means, the slope is 9.4.

(c) Comparison of Goodness-of-Fit

```
mod_c <- lm(price ~ sqft, data = collegetown)
summary(mod_c)
```

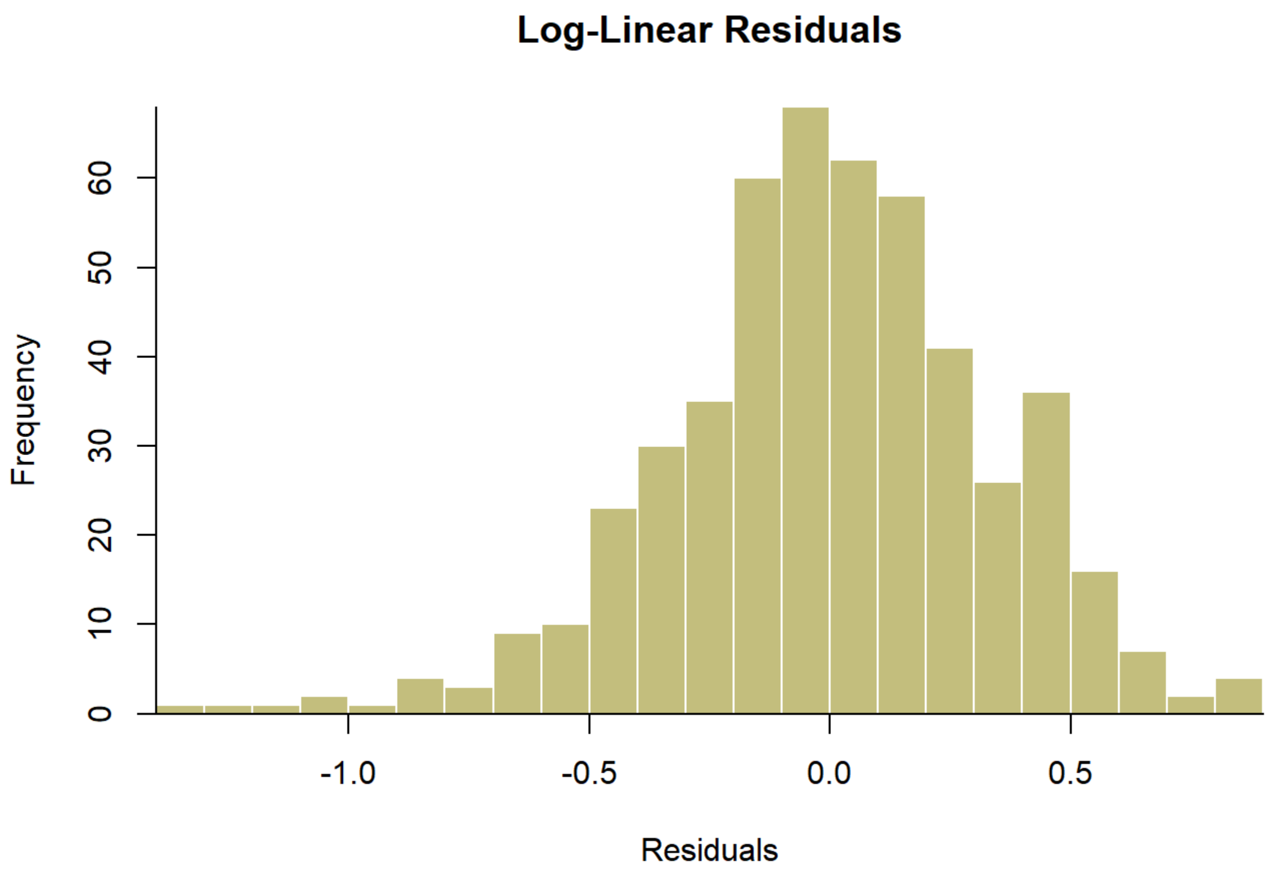
```
##
## Call:
## lm(formula = price ~ sqft, data = collegetown)
##
## Residuals:
##      Min       1Q  Median       3Q      Max
## -316.93  -58.90   -3.81   47.94  477.05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -115.4236    13.0882  -8.819   <2e-16 ***
## sqft          13.4029     0.4492  29.840   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 102.8 on 498 degrees of freedom
## Multiple R-squared:  0.6413, Adjusted R-squared:  0.6406
## F-statistic: 890.4 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
# Generalized R-squared for Log Models
y_true <- collegetown$price
gen_r2_a <- cor(y_true, exp(predict(mod_a)))^2
gen_r2_b <- cor(y_true, exp(predict(mod_b)))^2
r2_c <- summary(mod_c)$r.squared
```

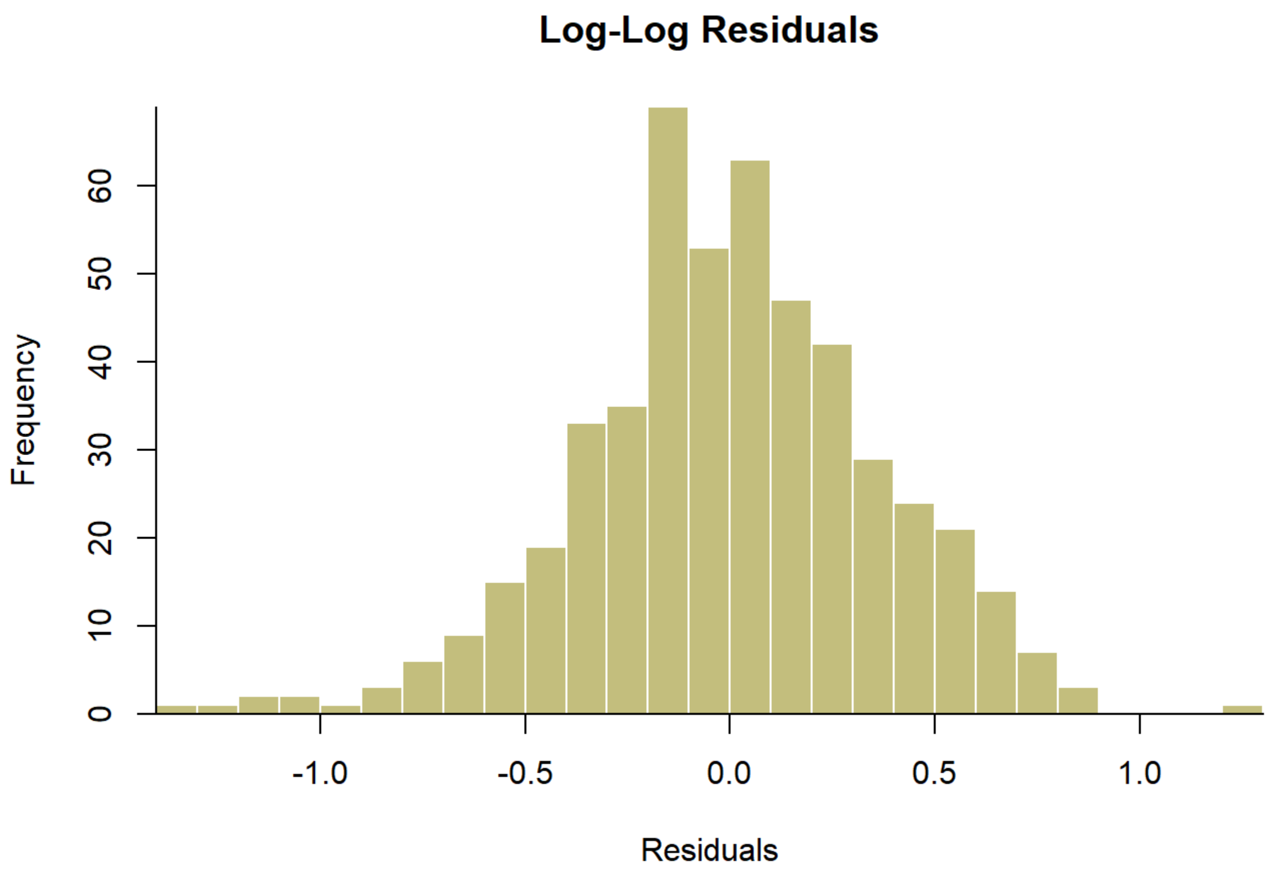
Answer: Comparing the generalized R^2, the Log-Linear model 0.6622 and Log-Log model 0.6445 both provide a better fit to the data than the simple linear model 0.6413.

(d) Normality of Residuals

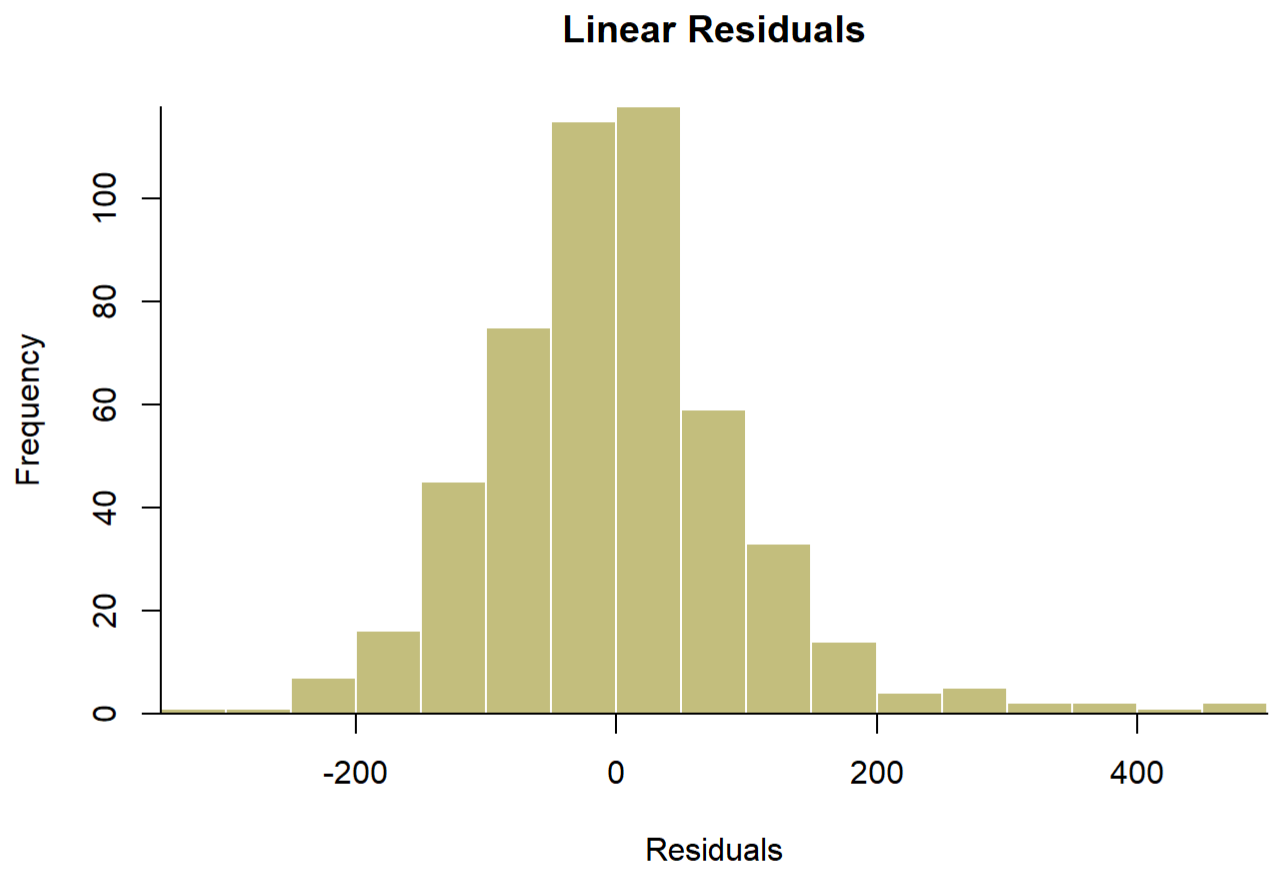
```
hist(residuals(mod_a),
     breaks = 20,
     col = "#c7bf7f",
     border = "white",
     main = "Log-Linear Residuals",
     xlab = "Residuals",
     xaxs = "i",
     yaxs = "i")
box(bty = "l")
```




```
hist(residuals(mod_b),
     breaks = 20,
     col = "#c7bf7f",
     border = "white",
     main = "Log-Log Residuals",
     xlab = "Residuals",
     xaxs = "i",
     yaxs = "i")
box(bty = "l")
```




```
hist(residuals(mod_c),
     breaks = 25,
     col = "#c7bf7f",
     border = "white",
     main = "Linear Residuals",
     xlab = "Residuals",
     xaxs = "i",
     yaxs = "i")
box(bty = "l")
```

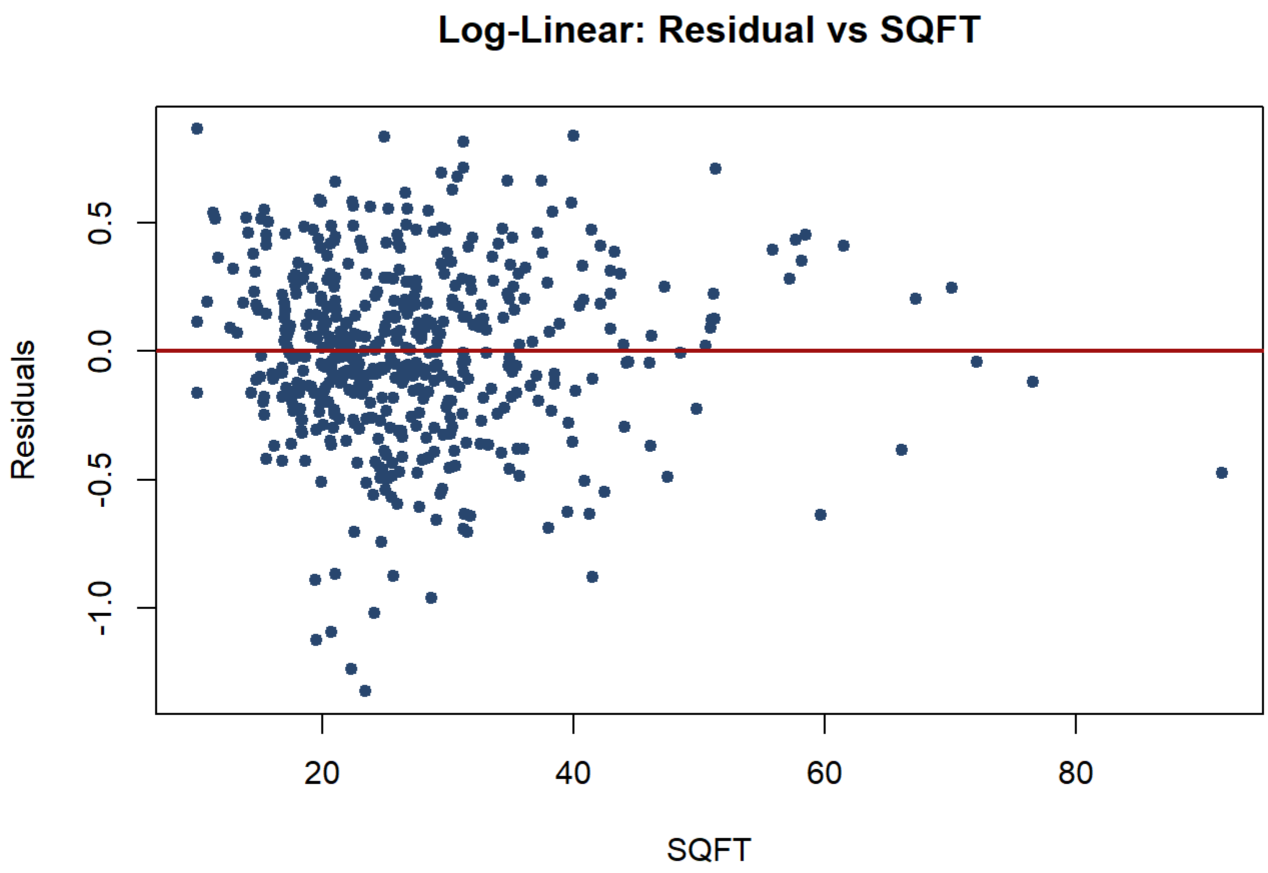



```
# Jarque-Bera Tests
jb_a <- jarque.bera.test(residuals(mod_a))
jb_b <- jarque.bera.test(residuals(mod_b))
jb_c <- jarque.bera.test(residuals(mod_c))
```

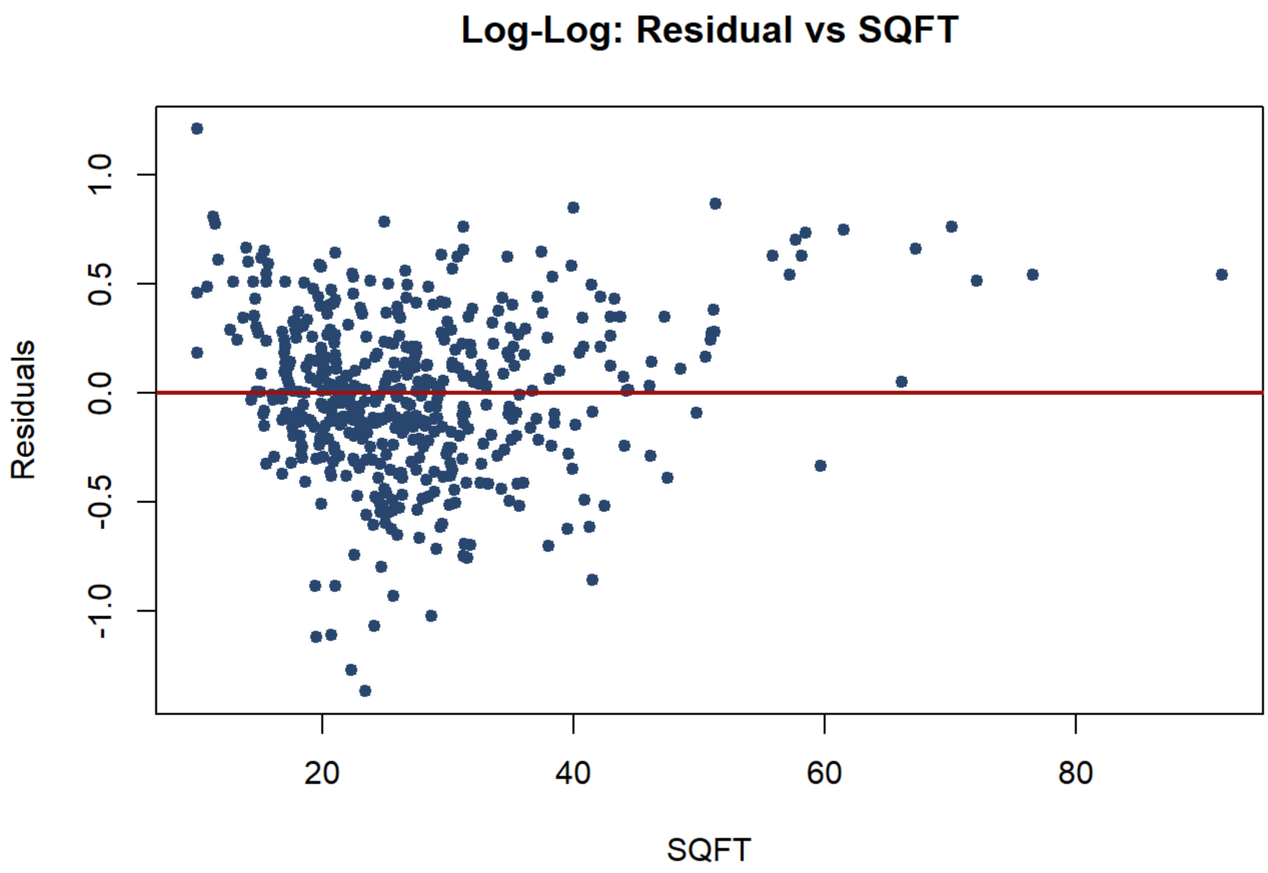
Answer: The Jarque-Bera test p-values are extremely low for all models, especially for the linear model ($p \approx 0$). This indicates that the residuals are not normally distributed. However, the log-transformed models show less extreme skewness in the histograms.

(e) Residual Patterns and Heteroscedasticity

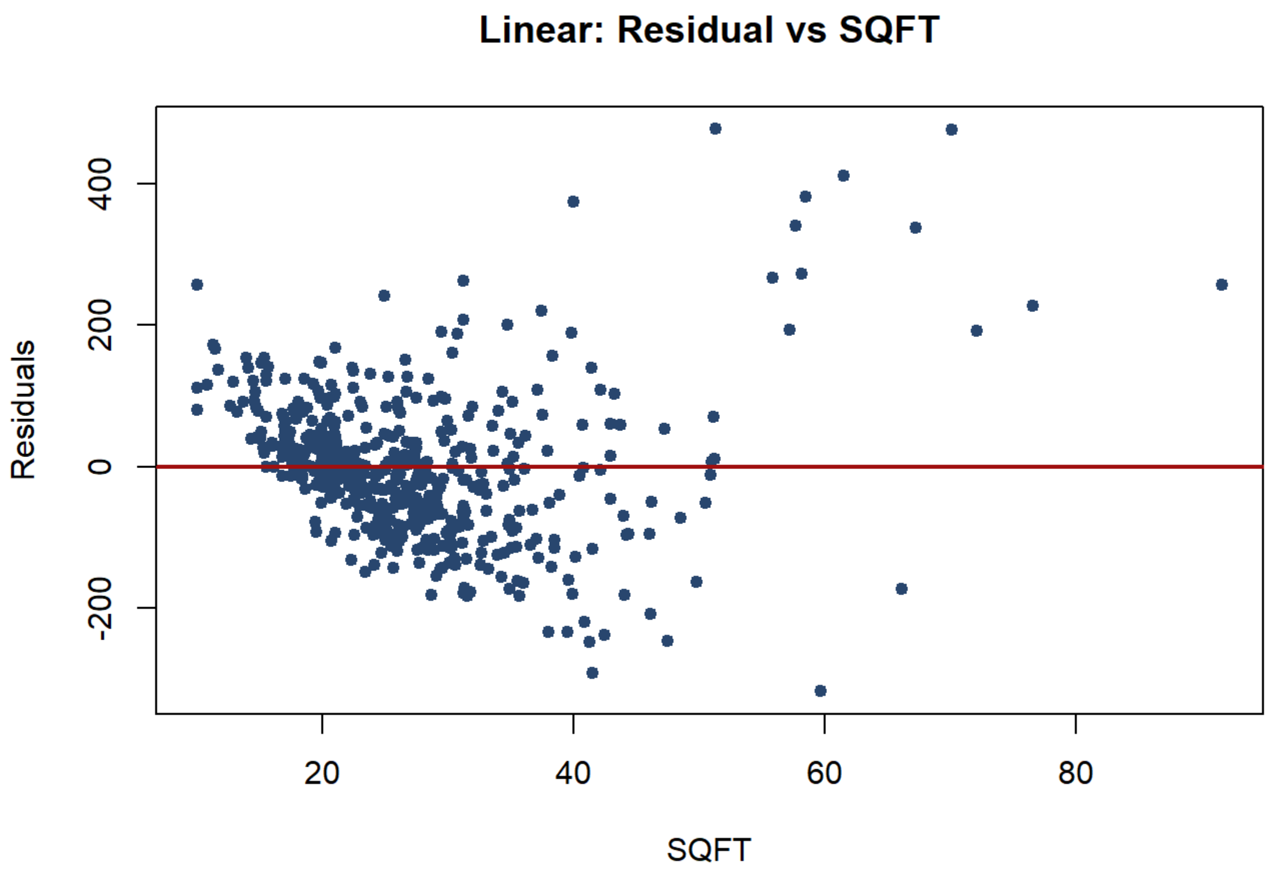
```
plot(collegetown$sqft, residuals(mod_a), main = "Log-Linear: Residual vs SQFT", xlab = "SQF
T", ylab = "Residuals"
     ,pch = 19, col = "#2b4a6f", cex = 0.8)
abline(h = 0, col = "#a21313", lwd = 2)
```




```
plot(collegetown$sqft, residuals(mod_b), main = "Log-Log: Residual vs SQFT", xlab = "SQFT", y
lab = "Residuals"
     ,pch = 19, col = "#2b4a6f", cex = 0.8)
abline(h = 0, col = "#a21313", lwd = 2)
```




```
plot(collegetown$sqft, residuals(mod_c), main = "Linear: Residual vs SQFT", xlab = "SQFT", yl
ab = "Residuals"
     ,pch = 19, col = "#2b4a6f", cex = 0.8)
abline(h = 0, col = "#a21313", lwd = 2)
```



Answer: The linear model's residuals show a distinct fan shape, indicating heteroscedasticity. The log-transformed models (a) and (b) help stabilize this variance, though some patterns may still remain.

(f)(g) Prediction for a 2700 sqft House

```
new_data <- data.frame(sqft = 27) # 2700 sqft corresponds to sqft = 27 in this dataset

# Predictions (Exp-transformed for Log models)
pred_a_log <- predict(mod_a, new_data, interval = "prediction", level = 0.95)
pred_a <- exp(pred_a_log)

pred_b_log <- predict(mod_b, new_data, interval = "prediction", level = 0.95)
pred_b <- exp(pred_b_log)

pred_c <- predict(mod_c, new_data, interval = "prediction", level = 0.95)

# Summary Table
results_comparison <- data.frame(
  Model = c("Log-Linear", "Log-Log", "Linear"),
  R_squared = c(gen_r2_a, gen_r2_b, r2_c),
  JB_p_value = c(jb_a$p.value, jb_b$p.value, jb_c$p.value),
  Pred_Price_2700 = c(pred_a[1], pred_b[1], pred_c[1])
)

kable(results_comparison, caption = "Comparison of Model Results and Predictions", digits =
4)
```

Comparison of Model Results and Predictions

Model	R_squared	JB_p_value	Pred_Price_2700
Log-Linear	0.6622	0e+00	214.2336
Log-Log	0.6445	8e-04	227.5386
Linear	0.6413	0e+00	246.4557

(h) Conclusion on Functional Form

Answer: Based on the analysis, the Log-Log model (b) is generally preferred. It addresses the non-linearity between house size and price more effectively than the linear model, yields a higher generalized R^2, and provides a more intuitive interpretation of the relationship through constant elasticity. Furthermore, it helps mitigate the heteroscedasticity observed in the original price-level residuals.